

CAKE Web App Guidance

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Continuous Addition Kinetic Elucidation (CAKE) is a kinetic method for determining catalyst and reactant order, rate constant, and poisoning from a single experiment.¹ The CAKE Web App has been built to simplify data processing and the calculation of these parameters. This uses code which is freely available to download and modify from <https://github.com/peterjhw07/CAKE>.

Guidance for how to use the CAKE Web App is given below. Essential items are marked in red and with an asterisk*. An example of CAKE Web App use is also described below (under Demonstration).

Guidance

First, one must fill in the parameters as described below.

*Upload

Upload: upload of reaction data.

- ***Select Data:** upload reaction data in Excel format (.xlsx). The first row is assumed to be headings, i.e., only data from the second row onward are used for fitting. Number of rows must be equal for all data.
- **Sheet Name:** specify the sheet name. Default is Sheet1.
- ***Time Column:** specify which column is the time column. The first column is 1.
- ***Reactant/Product Columns:** specify which column contains reactant/product data. The first column is 1. Reactant column, product column or both must be specified for “Fit Aspect” (see below) reactant, product or reactant + product respectively.
- **Fit Aspect:** specify whether the program should fit to the reactant, product or both. Default is reactant.

*Rxn Data

Reaction data: required reaction conditions.

- **Reactant Coefficient:** specifies the stoichiometric coefficient of reactant. Default is 1.
- **Product Coefficient:** specifies the stoichiometric coefficient of product. Default is 1.
- ***Catalyst Concentration:** specify the concentration of the catalyst solution being injected. **Uses units which achieve a value within a couple of orders of magnitude of 1**, as this significantly improves the fitting. For example, 5×10^{-5} M should be given as 50 μ M.
- ***Injection Rate:** specifies the rate of injection of the catalyst solution. **Uses same volume units as for “Catalyst Concentration” and the same time unit as for “Time Column”**.
- ***Initial Rxn Solution Volume:** specifies the initial volume of the reaction solution to which catalyst solution is being injected. **Uses same volume units as for “Catalyst Concentration”**.
- **Time of Injection:** specifies the time as which injection begins. **Uses the time unit of “Time Column”**. Default is 0, i.e., injection commences immediately.

Data Manip

Data manipulation: means to manipulate the reaction data.

Especially useful for noisy data and data not given as concentrations, e.g. data given in intensities from UV-Vis spectroscopy, NMR spectroscopy, IR spectroscopy, MS, etc.

- **Starting Reactant Concentration:** scales data to the Starting Reactant Concentration across the specified number of “Average Points” (see below). **Uses same concentration units as for “Catalyst Concentration”.**
- **Starting Product Concentration:** scales data to the Starting Reactant Concentration across the specified number of “Average Points” (see below). **Uses same units as for “Catalyst Concentration”.** Default is 0.0.
- **Final Product Concentration:** scales data to the Final Product Concentration across the specified number of “Average Points” (see below). **Uses same units as for “Catalyst Concentration”.**
- **Average Points:** specifies how many data points to average across for scaling to species concentration. Default is 0, i.e., no scaling.
- **Smoothing Window:** specifies number of data points for rolling average. Default is 1, i.e., no smoothing.

Advanced

Advanced: allows overriding of default fitting parameters.

May be required if fitting is especially poor (“Rate Constant” parameters below may need editing); no poisoning is desired (fix below “Start Time” to 0); reaction likely has unusual reaction orders (edit “Reactant Order” and “Catalyst Order” below) or very few data points are provided (increase “Interpolation Multiplier” below).

- **Est Rate Constant:** specifies starting value for fitting of the rate constant if “Min Rate Constant” and “Max Rate Constant” (see below) are used, or specifies exact rate constant if “Min Rate Constant” and “Max Rate Constant” (see below) are left empty. **Uses same concentration units as for “Catalyst Concentration” and same time unit as “Time Column”.** Default is for the starting value to be estimated.
- **Min Rate Constant:** specifies the lower bound for fitting of the rate constant. **Uses same concentration units as for “Catalyst Concentration” and same time unit as “Time Column”.** Fixes rate constant value if left empty. Default is the starting value $\times 1 \times 10^{-3}$.
- **Max Rate Constant:** specifies the upper bound for fitting of the rate constant. **Uses same concentration units as for “Catalyst Concentration” and same time unit as “Time Column”.** Fixes rate constant value if left empty. Default is the starting value $\times 1 \times 10^3$.
- **Est Start Time:** specifies starting value for fitting of poisoning delay if “Min Start Time” and “Max Start Time” (see below) are used, or specifies exact poisoning delay if “Min Start Time” and “Max Start Time” (see below) are left empty. **Uses same time unit as “Time Column”.** Default is 0.
- **Min Start Time:** specifies the lower bound for fitting of poisoning delay. Fixes start time value if left empty. **Uses same time unit as “Time Column”.** Default is 0.
- **Max Start Time:** specifies the upper bound for fitting of the poisoning delay. Fixes start time value if left empty. **Uses same time unit as “Time Column”.** Default is the final time value in the “Time Column”.

- **Est Reactant Order:** specifies starting value for fitting of reactant order if “Min Reactant Order” and “Max Reactant Order” (see below) are used, or specifies exact reactant order if “Min Reactant Order” and “Max Reactant Order” (see below) are left empty. Default is 1.
- **Min Reactant Order:** specifies the lower bound for fitting of reactant order. Fixes Reactant Order value if left empty. Default is 0.
- **Max Reactant Order:** specifies the upper bound for fitting of the poisoning delay. Fixes Reactant Order value if left empty. Default is 2.
- **Est Catalyst Order:** specifies starting value for fitting of Catalyst order if “Min Catalyst Order” and “Max Catalyst Order” (see below) are used, or specifies exact Catalyst order if “Min Catalyst Order” and “Max Catalyst Order” (see below) are left empty. Default is 1.
- **Min Catalyst Order:** specifies the lower bound for fitting of Catalyst order. Fixes Catalyst Order value if left empty. Default is 0.
- **Max Catalyst Order:** specifies the upper bound for fitting of the poisoning delay. Fixes Catalyst Order value if left empty. Default is 2.
- **Interpolation Multiplier:** specifies number of additional time points used for more accurate simulation. Default is 1, i.e., only time points provided in “Time Column” are used for simulation. Increase recommended for data sets with few data points (<25).

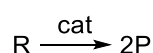
Output

- Once the above parameters have been filled in, the data can be fitted and/or downloaded, using “Fit” or “Download Fit” (see below). This changes the tab to the “Output” tab, eventually revealing results.
- **Fit:** yields results online, shown as the obtained fitted parameters and figures showing the fit upon the inputted data. These figures can be downloaded in SVG, PNG, EPS and PDF formats using the drop down format button and download button found below the figures.
- **Download Fit:** The downloaded fit outputs the inputted data and the obtained fit data (under FitData tab), the obtained fit parameters (under Outputs tab) and all the inputted parameters (under InputParams tab). This allows for easy tabulation of results and plotting of data in the desired software.
- Fitting typically takes ~5-60 s, depending on the amount of data, chosen fitting parameters and local internet speed.

Demonstration

An example fitting is shown for simulated reaction data. These simulated reaction data are available as “CAKE_example_data.xlsx”.

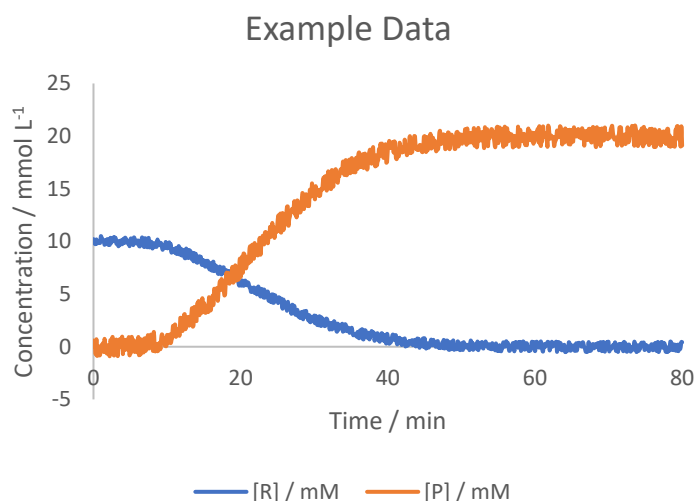
For a decomposition reaction of reactant “R” forming two product molecules “P”, catalysed by catalyst “cat”, that is first order in reactant and catalyst:



Data were simulated using the following parameters: reactant stoichiometric coefficient = 1; product stoichiometric coefficient = 2; reactant order = 1; catalyst order = 1; rate constant = $8.50 \times 10^{-2} \text{ mmol L}^{-1} \text{ min}^{-1}$; time of injection = 1.0 min; poisoning delay = 4.0 min; initial

reactant concentration = 10 mmol L⁻¹; initial product concentration = 0 mmol L⁻¹; final product concentration = 20 mmol L⁻¹; initial reaction volume = 10 mL or 1×10⁻² L; catalyst solution concentration = 50 mmol L⁻¹; rate of catalyst solution injection = 10 μL min⁻¹ or 1×10⁻⁵ L min⁻¹; a reactant noise factor which yielded $[R] = [R]_{\text{corr}} \pm [R]_0 \times \text{random}(0-0.05)$ and a product noise factor which yielded $[P] = [P]_{\text{corr}} \pm [P]_0 \times \text{random}(0-0.10)$.

	A	B	C
1	Time / min	[R] / mM	[P] / mM
2	0	10.22328	0.59667
3	0.1	9.952461	0.006834
4	0.2	9.755715	0.266997
5	0.3	9.760059	-0.7257
6	0.4	10.19514	0.715709
7	0.5	9.812772	-0.87259
8	0.6	9.999432	0.022112
9	0.7	9.884903	0.008014
10	0.8	9.8888	-0.80895
11	0.9	9.850461	-0.76568
12	1	10.49387	-0.10541
13	1.1	10.37282	0.158292
14	1.2	10.17935	0.077213
15	1.3	10.16717	0.323175
16	1.4	10.0993	-0.49982
17	1.5	9.676149	0.753005
18	1.6	9.905986	0.223811
19	1.7	10.10249	0.076301
20	1.8	10.08169	-0.62746
21	1.9	10.20881	-0.62353



These data were input into the program as shown below. Units are shown in brackets for clarity but were not put into the program.

Upload

- **Select Data:** CAKE_example_data.xlsx.
- **Sheet Name:** reaction_monitor.
- **Time Column:** 1.
- **Reactant Column:** 2.
- **Product Columns:** 3.
- **Fit Aspect:** “Reactant”.

Upload

Rxn Data

Data Manip

Advanced

Output

Select Data

Choose file

CAKE_example_data.xlsx

Sheet Name

reaction_monitor

Time Column

Reactant Column

Product Column

1

2

3

Fit Aspect

Reactant

Fit

Download Fit

Rxn Data

- **Reactant Coefficient:** 1 (default).
- **Product Coefficient:** 2.
- **Catalyst Concentration:** 50 (mmol L⁻¹).
- **Injection Rate:** 1E-5 (L min⁻¹).
- **Initial Rxn Solution Volume:** 1E-2 (L min⁻¹).
- **Time of Injection:** 1.0 (min).

Upload Rxn Data Data Manip Advanced Output

Reactant Coefficient: 1

Product Coefficient: 2

Catalyst Concentration: 50

Injection Rate: 1E-5

Initial Rxn Solution Volume: 1E-2

Time of Injection: 1.0

Fit Download Fit

Data Manip

Data are noisy so data manipulation was used.

- **Starting Reactant Concentration:** left blank (default). Could also use 10 (mmol L⁻¹).
- **Starting Product Concentration:** 0.0 (default).
- **Final Product Concentration:** left blank (default). Could also use 20 (mmol L⁻¹).
- **Average Points:** 5.
- **Smoothing Window:** 1 (default, i.e., no smoothing).

Upload Rxn Data Data Manip Advanced Output

Starting Reactant Concentration:

Starting Product Concentration: 0.0

Final Product Concentration:

Average Points: 5

Smoothing Window: 1

Fit Download Fit

Advanced

Not used - all left at defaults.

CAKE Web App Guidance

Upload	Rxn Data	Data Manip	Advanced	Output
Est Rate Constant	Min Rate Constant	Max Rate Constant		
Estimate	Estimate * 1E-3	Estimate * 1E3		
Est Start Time	Min Start Time	Max Start Time		
0	0	Rxn End Time		
Est Reactant Order	Min Reactant Order	Max Reactant Order		
1	0	2		
Est Catalyst Order	Min Catalyst Order	Max Catalyst Order		
1	0	2		
Interpolation Multiplier				
1				
Fit		Download Fit		

Output

Fit:

[Upload](#)
[Rxn Data](#)
[Data Manip](#)
[Advanced](#)
[Output](#)

	Rate Constant (k)	Reactant Order
Opt. Values	8.726587E-02	0.976223
Est. Error +/-	8.672398E-03	0.054705

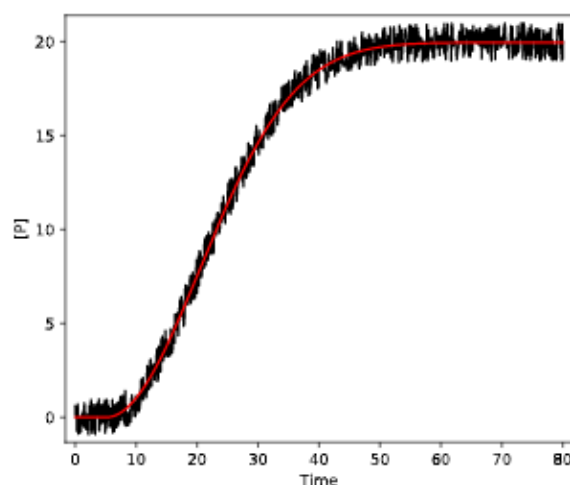
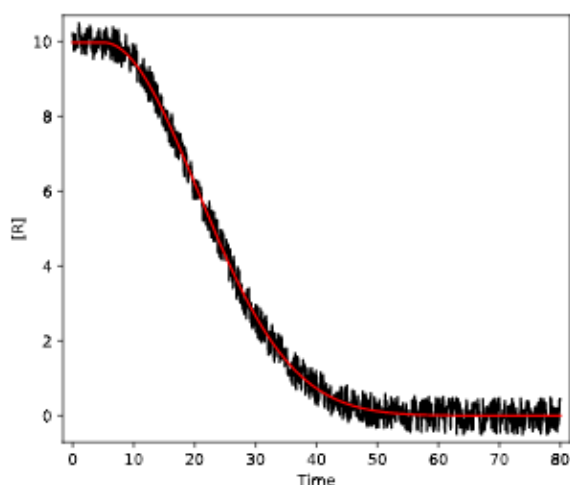
	Catalyst Order	Start Time
Opt. Values	0.977658	3.995205
Est. Error +/-	0.127095	0.697669

Residual Sum of Squares for Optimization: 339.938564.

R² Value of Fit: 0.996748.

Catalyst Poisoning (if applicable): 1.997603E-01
 Catalyst Poisoning Error: 0.034883473980219716

[Fit](#)
[Download Fit](#)



SVG



Download Fit:

These results are available as "CAKE_example_fitting.xlsx".

- FitData:

	A	B	C	D	E	F	G	H	I
1		Time / min	[R] / mM	[P] / mM	Transform	Transform	Transform	Fit R	Fit P
2	0	0	10.22328	0.59667	0	10.22328	0.59667	9.977332	0
3	1	0.1	9.952461	0.006834	0.1	9.952461	0.006834	9.977332	0
4	2	0.2	9.755715	0.266997	0.2	9.755715	0.266997	9.977332	0
5	3	0.3	9.760059	-0.7257	0.3	9.760059	-0.7257	9.977332	0
6	4	0.4	10.19514	0.715709	0.4	10.19514	0.715709	9.977332	0
7	5	0.5	9.812772	-0.87259	0.5	9.812772	-0.87259	9.977332	0
8	6	0.6	9.999432	0.022112	0.6	9.999432	0.022112	9.977332	0
9	7	0.7	9.884903	0.008014	0.7	9.884903	0.008014	9.977332	0
10	8	0.8	9.8888	-0.80895	0.8	9.8888	-0.80895	9.977332	0
11	9	0.9	9.850461	-0.76568	0.9	9.850461	-0.76568	9.977332	0
12	10	1	10.49387	-0.10541	1	10.49387	-0.10541	9.977332	0
13	11	1.1	10.37282	0.158292	1.1	10.37282	0.158292	9.977332	0
14	12	1.2	10.17935	0.077213	1.2	10.17935	0.077213	9.977332	0
15	13	1.3	10.16717	0.323175	1.3	10.16717	0.323175	9.977332	0
16	14	1.4	10.0993	-0.49982	1.4	10.0993	-0.49982	9.977332	0
17	15	1.5	9.676149	0.753005	1.5	9.676149	0.753005	9.977332	0
18	16	1.6	9.905986	0.223811	1.6	9.905986	0.223811	9.977332	0
19	17	1.7	10.10249	0.076301	1.7	10.10249	0.076301	9.977332	0
20	18	1.8	10.08169	-0.62746	1.8	10.08169	-0.62746	9.977332	0
21	19	1.9	10.20881	-0.62353	1.9	10.20881	-0.62353	9.977332	0
22	20	2	9.676498	0.145266	2	9.676498	0.145266	9.977332	0
23	21	2.1	10.09532	-0.33756	2.1	10.09532	-0.33756	9.977332	0

- Outputs:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1		Rate Constant	Reactant Order	Catalyst Order	Zero Time	lyst Poison	Constant	React Order	lyst Order	ro Time	Erst Poisonin	RSS	R2
2	0	0.087266	0.976223	0.977658	3.995205	0.19976	0.008672	0.054705	0.127095	0.697669	0.034883	339.9386	0.996748
3													

- InputParams:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1		Stoich R	Stoich P	R 0	P 0	P_end	at Add Rat	Injection T	Ion Coun	R col	P col	Time col	Ion Calibra	athing Wir	plation N
2	0	1	2		0		0.05	1		1	2	0	5	1	
3															

N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC
ing Wir	plation	Mutting	Aspek	Estimate	Minimum	Maximum	Order Estim	der Min	der Max	Order Estim	der Min	der Max	Time Estim	Time Mini	Time Maximum
1		1 r					1	0	2	1	0	2	0	0	End Rxn Time

References

- 1 P. Williams, C. Killeen, I. Chagunda, B. Henderson, S. Donnecke, W. Munro, J. Sidhu, D. Kraft, D. Harrington and S. McIndoe, *Continuous Addition Kinetic Elucidation: Catalyst and Reactant Order, Rate Constant, and Poisoning from a Single Experiment*, Chemistry, 2022.